

triggered a cascade of selling that briefly overwhelmed the market.

The Flash Crash taught the financial world a crucial lesson: in algorithmic trading, data isn't just important—it's everything. The quality, timeliness, and proper handling of financial data determine whether an algorithm makes profitable trades or catastrophic mistakes.

In this chapter, we're diving into the world of financial data—the raw material that feeds every trading algorithm. We'll explore different types of financial data, learn how to access this data using Python libraries, and discover essential techniques for cleaning and processing data before feeding it to our algorithms.

By the end of this chapter, you'll understand how to acquire, clean, and prepare financial data—critical skills that form the foundation for developing effective trading algorithms. Let's begin by exploring the fascinating landscape of financial data.

Understanding Financial Data: The Markets in Numbers

Financial data is simply numerical information about markets and securities. But beneath this simple definition lies a rich ecosystem of data types, formats, and sources that algorithmic traders use to make decisions.

Stock Prices: The Fundamentals

Stock price data is the most common type of financial data and forms the backbone of most trading strategies. When we talk about stock prices, we're usually referring to several key metrics:

- **Open:** The price at which a stock begins trading when the market opens
- **High:** The highest price the stock reaches during the trading session